

Verifying an agentic GIS routing workflow

Fiber route comparison on the shortened Mombasa corridor: baseline, roads, power and rail

Justin Rogers

September 28, 2026

GGS 662: Agentic GeoAI Assignment 2

2026-09-27

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GGS 662: Agentic GeoAI Assignment 2

About 15 s. Last week an agent built a fiber-routing workflow for the shortened Mombasa corridor and produced routes, a map and a cost table. This week the question is not whether it ran, but whether the results are correct and defensible. I will give the agent framework briefly, then spend most of the time on the checks and what they found.

Agent framework: six dimensions (04_01)

Goal

Compare baseline, roads, power and rail routes between the exact route1_short endpoints (east = start); report length and USD 10/m cost.

Agency

Agent cleans, nodes, repairs gaps and routes. **Human** fixed endpoints, 50 m gap limit, 500 m endpoint limit and the rate.

Environment

Saved 03_01 GeoPackages, AGENTS.md rules, 3,000 m study rectangle, measured in EPSG:32737. No new downloads.

Feedback

Stop with endpoint_too_far, no_path or empty_network. A failed check means revise, never relax a threshold.

Tools

geopandas, shapely, networkx (Dijkstra), pandas; matplotlib + contextily for maps.

Verification

Independent audit script, synthetic cases, controlled reruns. Real-world claims need imagery or authoritative data.

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Agent framework: six dimensions (04_01)

About 40 s. Goal: four candidate routes between the exact endpoints of the shortened baseline, easterly point as start, with length and a classroom cost of 10 dollars per metre. Environment: only the saved 03_01 files and the AGENTS.md rules. Tools: geopandas, shapely, networkx. Agency: the agent chooses how to clean, node, repair and route, but a human set the endpoints, the 50 m gap limit, the 500 m endpoint limit and the rate. Feedback: stop with a named status rather than invent a route; a failed check means revise, not raise a threshold. Verification: an audit that recomputes everything from saved files, plus outside evidence for real-world claims.

Agent framework: six dimensions (04_01)		
Goal Compare baseline, roads, power and rail routes between the exact route1_short endpoints (east = start); report length and USD 10/m cost.	Environment Saved 03_01 GeoPackages, AGENTS.md rules, 3,000m study rectangle, measured in EPSG:32737. No new downloads.	Tools geopandas, shapely, networkx (Dijkstra), pandas, matplotlib + contextily for maps.
Agency Agent cleans, nodes, repairs gaps and routes. Human fixed endpoints, 50m gap limit, 500m endpoint limit and the rate.	Feedback Stop with endpoint_too_far, no_path or empty_network. A failed check means revise, never relax a threshold.	Verification Independent audit script, synthetic cases, controlled reruns. Real-world claims need imagery or authoritative data.

Stays with a human

Whether an inferred connector or a 2D crossing is actually buildable, and which corridor to pursue.

Access rights, permits and engineering feasibility are not in the data, so the agent cannot establish them.

Feedback that changes the next action

The audit flags the rail route switching lines at a bridge (−3.99374, 39.54626) with no shared OSM node.

If imagery shows the lines are grade-separated: **remove that transfer and reroute rail**, then report the new length or no_path. Do not raise a threshold to keep the old answer.

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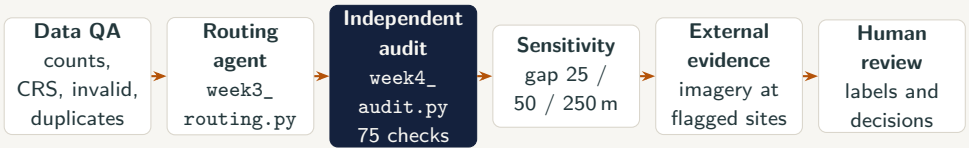
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Where the human decides and where the agent revises

About 25 s. One decision that must stay human: whether anything the model inferred, a connector or a crossing, can actually be built, and which corridor to pursue; that depends on access rights and engineering that are not in the data. One piece of feedback that should change the agent’s next action: the audit found the rail route switching onto a bridge where the two lines share no OSM node. If imagery confirms they are at different levels, the agent should delete that transfer and reroute, accepting a longer route or no_path, rather than loosening a threshold.

Stays with a human	Feedback that changes the next action
Whether an inferred connector or a 2D crossing is actually buildable, and which corridor to pursue. Access rights, permits and engineering feasibility are not in the data, so the agent cannot establish them.	The audit flags the rail route switching lines at a bridge (−3.99374, 39.54626) with no shared OSM node. If imagery shows the lines are grade-separated: remove that transfer and reroute rail , then report the new length or no_path. Do not raise a threshold to keep the old answer.

Approach: checks designed around the agent, not after it



Independent of the code. The audit never imports the routing script; it recomputes every value from the saved files and the 03_01 inputs.

Not independent of the author. The same AI tool wrote both scripts, so the audit verifies computations. Real-world claims need outside evidence.

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Verifying an agentic GIS routing workflow

Approach: checks designed around the agent, not after it

About 25 s. I followed the 04_02 validation stack: data QA, the routing agent, then a separate audit script that imports no routing code and rebuilds every length, cost and connector from the saved files; controlled reruns that change only the gap tolerance; outside evidence at the sites the audit flags; and a human applies the labels. Caveat: the same AI tool wrote both scripts, so the audit is verification, not independent validation.

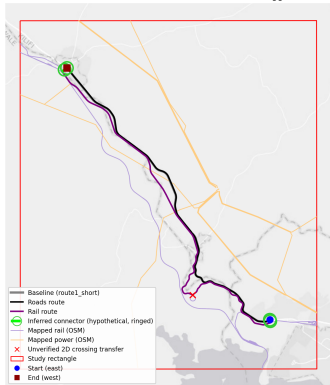
Approach: checks designed around the agent, not after it

Independent of the code. The audit never imports the routing script; it recomputes every value from the saved files and the 03_01 inputs.

Not independent of the author. The same AI tool wrote both scripts, so the audit verifies computations. Real-world claims need outside evidence.

Route map: what the agent built and what it inferred

Week 4 audit: shortened corridor routes, connectors and flagged crossings



Route and infrastructure data © OpenStreetMap contributors (ODbL). Basemap: Esri World Light Gray Canvas (Esri, HERE, Garmin, © OpenStreetMap contributors).

Green rings: inferred connectors. Roads 0.59 m and 0.72 m; rail 194.8 m and 372.7 m. All four are endpoint connectors: hypothetical new construction.

Red cross: flagged crossing. The rail route switches onto a bridge-tagged line where the two lines share no OSM vertex.

Lengths. Baseline 22,896.8 m; roads 22,897.6 m; rail 25,902.4 m. Power: endpoint_too_far.

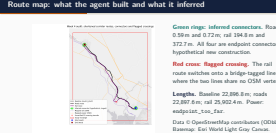
Data © OpenStreetMap contributors (ODbL). Basemap: Esri World Light Gray Canvas.

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Route map: what the agent built and what it inferred

About 35 s. Grey baseline and black roads route overlap almost everywhere, under one metre apart in total length. Purple is rail, 3 km longer. Green rings mark every place the model invented geometry: two sub-metre snaps for roads, and 195 m and 373 m links for rail, all endpoint connectors, drawn separately from mapped infrastructure. The red cross is where the rail route jumps onto a bridge at a point with no shared node, the one failed check. Power has no route: its start is 1.8 km from any mapped power line.



Verification evidence: 75 reproducible checks

71

PASS

1

FAIL: 2D crossing at a bridge

3

NOT_CHECKED: external evidence

Category	PASS	FAIL	NOT_CHK	What it checks
Provenance and data QA	7	0	0	counts, CRS, invalid geometry, duplicates
Comparison table and cost	9	0	0	four rows, blank unavailable values, length × USD 10/m
Route geometry and length	23	0	0	endpoints, continuity, metres, edge sum within 0.01 m
Connectors	6	0	0	limits, containment, counted once
Unavailable route (power)	1	0	0	endpoint distance recomputed
Synthetic cases	20	0	0	6 cases × 3 runs, plus the 230 m oracle
Sensitivity	5	0	0	only the gap setting changed
Spatial assumption	0	1	0	route transfers only at OSM junctions
External evidence	0	0	3	imagery at flagged sites (pending)

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Verification evidence: 75 reproducible checks

About 35 s. Each check records method, expected, observed, evidence file and PASS, FAIL or NOT_CHECKED; missing evidence never counts as a pass. The computational checks all pass: exact endpoints, geometry length equal to the traversed-edge sum within 1 cm, metres not degrees (UTM within 0.036% of geodesic), cost equals length times 10, connectors counted once. Synthetic cases cover the five AGENTS.md situations plus connector-counted-once in all three runs, and a hand oracle: 80 + 120 + 30 m gap = 230 m and USD 2,300. One fail and three not-checked are the interesting ones.

Verification evidence: 75 reproducible checks				
	71 PASS	1 FAIL: 2D crossing at a bridge	3 NOT_CHECKED: external evidence	
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External evidence	0	0	3	imagery at flagged sites (pending)

What the checks caught

FAIL: a 2D crossing at a rail bridge

The route moves from way 891041250 onto bridge-tagged way 239559205 at (−3.993745, 39.54626); neither line has a vertex there. A modelling assumption, not a proven error: needs imagery.

A bug in the agent’s own checker

The connector-limit check hard-coded 50 m, so the 250 m rerun was checked against the wrong limit. It now reads the run’s setting.

A wrong claim in the Week 3 answers

The write-up said rail used a gap connector. `connectors.gpkg` shows all four used connectors are endpoint connectors (rail: 194.8 m and 372.7 m). Corrected.

A justified failure: power

`endpoint_too_far` confirmed independently: start 1,839.2 m and end 995.1 m from the nearest mapped power line, against a 500 m limit.

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What the checks caught

About 45 s. First, the one failure: the audit looks for places where the route changes from one OSM line to another at a point that is not a shared vertex, which is exactly where "2D crossings connect" can be wrong. It found one, on a bridge-tagged rail line. Second, the agent's own Week 3 discussion claimed rail used a gap connector; the saved connectors file shows all four are endpoint connectors, so the prose was wrong though the numbers were right. Third, the built-in check hard-coded 50 m and would have mis-checked the 250 m rerun. Fourth, a justified no-route result: power really is 1.8 km from the start. A successful run alone would have hidden the first three.

What the checks caught	What the checks caught
FAIL: a 2D crossing at a rail bridge The route moves from way 891041250 onto bridge-tagged way 239559205 at (−3.993745, 39.54626); neither line has a vertex there. A modelling assumption, not a proven error: needs imagery.	A wrong claim in the Week 3 answers The write-up said rail used a gap connector. <code>connectors.gpkg</code> shows all four used connectors are endpoint connectors (rail: 194.8 m and 372.7 m). Corrected.
A bug in the agent's own checker The connector-limit check hard-coded 50 m, so the 250 m rerun was checked against the wrong limit. It now reads the run's setting.	A justified failure: power <code>endpoint_too_far</code> confirmed independently: start 1,839.2 m and end 995.1 m from the nearest mapped power line, against a 500 m limit.

Sensitivity: changing only the gap tolerance changes no route outcome

Gap tolerance	Roads	Power	Rail	Power parts*	Ranking
25 m	ok, 22,897.6 m	endpoint_too_far	ok, 25,902.4 m	6	baseline < roads < rail
50 m (original)	ok, 22,897.6 m	endpoint_too_far	ok, 25,902.4 m	3	baseline < roads < rail
250 m	ok, 22,897.6 m	endpoint_too_far	ok, 25,902.4 m	1	baseline < roads < rail

*Power connected components after gap repair (9 before). Runs: sensitivity_gap_25m, default, gap_tolerance_250m.

Why nothing moves. No route uses a gap connector, and power fails on the 1,839 m endpoint distance, which no gap setting can fix. Connector counts and lengths are identical in all three runs.

What this does not show. One setting at three values is not global robustness (Saltelli et al., 2019). A common rate of USD 5, 10 or 20/m scales costs but keeps the same order.

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Sensitivity: changing only the gap tolerance changes no route outcome

About 30 s. The rerun changes only the gap tolerance, and the audit confirms nothing else in the settings changed. At 25, 50 and 250 m every status, length, connector and the ranking are identical. Only the power layer’s internal fragmentation responds: 9 raw pieces become 6, 3 or 1. So the power failure is not a threshold artefact. Limit: a single-parameter rerun cannot show robustness to the endpoint limit or the crossing rule.

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Conclusions: what I trust and what stays unresolved

Conclusion	Label (04_02 §16)	Evidence
Lengths, endpoints, containment and costs are computed correctly	VERIFIED	R-*, C-*
Connectors within limits, counted once; all are endpoint connectors	VERIFIED	R-*-conn*
Power unavailable because its start is 1,839 m from mapped lines	VERIFIED; a real line there: UNRESOLVED	U-power
Ranking baseline < roads < rail holds for a 25–250 m gap tolerance	not SENSITIVE to this one setting	S-*
The rail transfer at the bridge is buildable	UNRESOLVED (check failed)	V-crossings
Connector sites and the crossing match conditions on the ground	TODO: PENDING imagery	E-*
Access rights, permits, feasibility and real costs	UNRESOLVED	none in data

TODO: Add your imagery screenshots of the 3 sites in external_evidence.csv, rerun week4_audit.py, then relabel the E-* row EXTERNALLY SUPPORTED or UNRESOLVED.

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Conclusions: what I trust and what stays unresolved

About 35 s. What I trust: the computations are verified, and the power failure is real under this data. The ranking does not depend on the gap tolerance. What I do not trust yet: the rail transfer at the bridge, which failed the crossing check, and whether the connector sites look the way the model assumes. **TODO: say what your imagery showed.** Nothing here establishes access rights, permits or real costs; proximity is not permission, and 10 dollars per metre is a classroom value. Verified, not yet validated.

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Data, tools and readings

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- Basemaps: Esri World Light Gray Canvas and World Street Map (Esri, HERE, Garmin, © OpenStreetMap contributors).
- GGS 662 notebooks 03_01, 03_02, 04_01, 04_02 and AGENTS.md. Python: geopandas, shapely, networkx, pandas, pyproj.
- Oreskes, Shrader-Frechette & Belitz (1994), *Science*, doi:10.1126/science.263.5147.641
- Haklay (2010), *Environment and Planning B*, doi:10.1068/b35097
- Saltelli et al. (2019), *Environmental Modelling & Software*, doi:10.1016/j.envsoft.2019.01.012

AI-use statement **TODO: EDIT IN YOUR OWN WORDS**

I used Claude Code (Anthropic, Claude Opus 5.5) to write the Week 3 routing script and the Week 4 audit script, to review both for errors, and to draft these slides. The audit caught a hard-coded connector limit and a wrong claim in the tool’s own Week 3 write-up.

I reviewed and ran every script, checked the flagged sites in imagery, and edited the slides. Because the same tool wrote the code and the audit, I treat the audit as verification, not independent validation.

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└ Sources and AI use

About 10 s. Data from OpenStreetMap contributors under ODbL, with Esri basemaps. Readings are the three from 04_02 that shaped the audit. **TODO: rewrite the AI statement in your own words and make sure every claim in it is true, including the imagery check.**

Sources and AI use

Data, tools and readings

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Supporting slides

Full check table, synthetic cases and evidence files

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Full check table, synthetic cases and evidence files

Supporting: per-route checks (all PASS)

Check (tolerance)	Baseline	Roads	Rail
Valid, nonempty single LineString	PASS	PASS	PASS
Endpoints match baseline (≤ 0.01 m)	0.0000 / 0.0000 m	0.0000 / 0.0000 m	0.0000 / 0.0000 m
No loops or self-crossing (<code>is_simple</code>)	True	True	True
Inside study rectangle	True	True	True
Geometry vs traversed-edge sum (≤ 0.01 m)	0.000000 m	0.009833 m	0.007000 m
Metres not degrees (UTM vs geodesic, $< 0.1\%$)	−0.036%	−0.036%	−0.036%
Not shorter than straight line	1.111×	1.111×	1.257×
Cost = length × USD 10/m	228,967.56	228,975.93	259,023.67
Connectors counted once (count, total)	0, 0 m	2, 1.3139 m	2, 567.5396 m
Each connector ≤ 500 m, inside, on route, at endpoint	–	0.59, 0.72 m	194.85, 372.69 m
Non-connector length on mapped lines (± 1 m)	–	22,896.4 m	25,334.9 m

Source: `validation_table.csv` (R-* checks). Note: roads edge-sum difference (0.0098 m) passes but sits close to the 0.01 m tolerance, consistent with 1 cm node rounding.

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No loops or self-crossing (<code>is_simple</code>)	True	True	True
Inside study rectangle	True	True	True
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Supporting: provenance, data QA and comparison table

Check ID	Observed	Status
P-settings	Settings match spec: gap 50 m, endpoint 500 m, margin 3,000 m, EPSG:32737	PASS
D-roads	4,742 features, EPSG:4326, 0 missing / empty / invalid, all intersect study area	PASS
D-power	15 features, 0 missing / empty / invalid	PASS
D-rail	77 features, 0 missing / empty / invalid	PASS
D-baseline	1 LineString, valid	PASS
D-settlements	30 features (29 points, 1 polygon), map context only	PASS
D-roads-dup	2,292 of 4,742 edges are reversed OSMnx duplicates; removed before routing	PASS
C-rows	baseline, roads, power, rail: one row each	PASS
C-cols	nine required CSV columns	PASS
C-base	baseline connector values 0, 0, 0	PASS
C-unavail	power: endpoint_too_far, blank length and cost (not zero)	PASS
C-gpkg	routes.gpkg holds baseline, roads, rail	PASS

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Verifying an agentic GIS routing workflow

Supporting: provenance, data QA and comparison table

Check ID	Observed	Status
P-settings	Settings match spec: gap 50 m, endpoint 500 m, margin 3,000 m, EPSG:32737	PASS
D-roads	4,742 features, EPSG:4326, 0 missing / empty / invalid, all intersect study area	PASS
D-power	15 features, 0 missing / empty / invalid	PASS
D-rail	77 features, 0 missing / empty / invalid	PASS
D-baseline	1 LineString, valid	PASS
D-settlements	30 features (29 points, 1 polygon), map context only	PASS
D-roads-dup	2,292 of 4,742 edges are reversed OSMnx duplicates; removed before routing	PASS
C-rows	baseline, roads, power, rail: one row each	PASS
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C-gpkg	routes.gpkg holds baseline, roads, rail	PASS

Supporting: unavailable route, spatial assumption, sensitivity, external evidence

Check ID	Observed	Status
C-rate	one common rate, USD 10/m	PASS
U-power	start 1,839.2 m, end 995.1 m from nearest power line (>500 m)	PASS
V-crossings	1 transfer at a non-shared vertex; involves a bridge-tagged way	FAIL
S-25m-ctrl	only “Gap connector limit” changed, 50 → 25 m	PASS
S-25m-out	no differences; ranking unchanged; power parts 6	PASS
S-250m-ctrl	only “Gap connector limit” changed, 50 → 250 m	PASS
S-250m-out	no differences; ranking unchanged; power parts 1	PASS
S-rate	same order at USD 5, 10, 20/m	PASS
E-rail connector 3	195 m connector site (−4.008849, 39.591099): no evidence yet	NOT_CHECKED
E-rail connector 4	373 m connector site (−3.864413, 39.472359): no evidence yet	NOT_CHECKED
E-rail crossing 1	crossing site (−3.993745, 39.54626): no evidence yet	NOT_CHECKED

TODO: Fill in external_evidence.csv (source, date, provenance, independent_of_osm, observation, supports_claim) and rerun the audit.

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S-25m-ctrl	only “Gap connector limit” changed, 50 → 25 m	PASS
S-25m-out	no differences; ranking unchanged; power parts 6	PASS
S-250m-ctrl	only “Gap connector limit” changed, 50 → 250 m	PASS
S-250m-out	no differences; ranking unchanged; power parts 1	PASS
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E-rail connector 3	195m connector site (−4.008849, 39.591099): no evidence yet	NOT_CHECKED
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Supporting: synthetic cases and known-answer oracle

Case (AGENTS.md)	Expected	50 m run	25 m run	250 m run
Already connected: components	1	1 PASS	1 PASS	1 PASS
Gap below threshold: components after repair	1	1 PASS	1 PASS	1 PASS
Gap above threshold: components after repair	2	2 PASS	2 PASS	2 PASS
Endpoint on a line interior: status	ok	ok PASS	ok PASS	ok PASS
Disconnected endpoints: status	no_path	no_path PASS	no_path PASS	no_path PASS
Connector counted once: route length (10 m + gap + 20 m)	= expected	55.0 m PASS	42.5 m PASS	155.0 m PASS
Oracle: 80 m + 120 m + 30 m gap at USD 10/m	230 m, USD 2,300	230 m, USD 2,300 PASS		
Oracle fault: gap double-counted	flagged	260 m $\neq$ 230 m flagged PASS		

The gap in the connector case is half the run's tolerance, so the expected length differs per run. Source: `routing_report.md` in each run folder; audit IDs T-*,

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Gap below threshold: components after repair	1	1 PASS	1 PASS	1 PASS
Gap above threshold: components after repair	2	2 PASS	2 PASS	2 PASS
Endpoint on a line interior: status	ok	ok PASS	ok PASS	ok PASS
Disconnected endpoints: status	no_path	no_path PASS	no_path PASS	no_path PASS
Connector counted once: route length (10 m + gap + 20 m)	= expected	55.0 m PASS	42.5 m PASS	155.0 m PASS
Oracle: 80 m + 120 m + 30 m gap at USD 10/m	230 m, USD 2,300	230 m, USD 2,300 PASS		
Oracle fault: gap double-counted	flagged	260 m $\neq$ 230 m flagged PASS		
The gap in the connector case is half the run's tolerance, so the expected length differs per run. Source: <code>routing_report.md</code> in each run folder; audit IDs T-*,				

Supporting: evidence files, run commands and versions

Run commands (every argument has a default)

- python Claude/03_Project/week3_routing.py → outputs/default/
- python week3_routing.py --gap-tolerance-m 25 --out outputs/sensitivity_gap_25m
- python week3_routing.py --gap-tolerance-m 250 --out outputs/gap_tolerance_250m
- python Claude/04_Project/week4_audit.py → outputs/week4_validation/

Evidence files

- Each run folder: route_comparison.csv, routes.gpkg, connectors.gpkg, route_map.png, routing_report.md
- Claude/04_Project/outputs/week4_validation/: validation_table.csv, validation_report.md, validation_map.png, inferred_crossings.gpkg, external_evidence.csv

Versions: Python 3.12.2; geopandas 1.1.4; shapely 2.1.2; pyproj 3.8.0; networkx 3.6.1; pandas 2.3.3.

Output location: 04_02 names outputs/week3_routing/ and outputs/week4_validation/ at the repo root; these are kept under Claude/ so the course repo inputs stay read-only.

2026-09-27

Verifying an agentic GIS routing workflow

Supporting: evidence files, run commands and versions

Supporting: evidence files, run commands and versions

Run commands (every argument has a default)

- python Claude/03_Project/week3_routing.py → outputs/default/
- python week3_routing.py --gap-tolerance-m 25 --out outputs/sensitivity_gap_25m
- python week3_routing.py --gap-tolerance-m 250 --out outputs/gap_tolerance_250m
- python Claude/04_Project/week4_audit.py → outputs/week4_validation/

Evidence files

- Each run folder: route_comparison.csv, routes.gpkg, connectors.gpkg, route_map.png, routing_report.md
- Claude/04_Project/outputs/week4_validation/: validation_table.csv, validation_report.md, validation_map.png, inferred_crossings.gpkg, external_evidence.csv

Versions: Python 3.12.2; geopandas 1.1.4; shapely 2.1.2; pyproj 3.8.0; networkx 3.6.1; pandas 2.3.3.

Output location

04_02 names outputs/week3_routing/ and outputs/week4_validation/ at the repo root; these are kept under Claude/ so the course repo inputs stay read-only.